

Testosterone Free, Profile I

Test Details

METHODOLOGY

Albumin: Colorimetric

Testosterone: Electrochemiluminescence immunoassay (ECLIA)

SHBG: Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Albumin; free testosterone, calculated; sex hormone binding globulin; testosterone by ECLIA

Use

Free testosterone is the concentration of unbound testosterone in serum.¹ The majority (approximately 60% to 90%) of serum total testosterone is associated with sex hormone binding globulin (SHBG); this fraction is tightly bound and biologically unavailable to its target tissues. The remaining bioavailable testosterone is mostly bound to albumin, with only a small fraction (approximately 0.5% to 2%) circulating in the free form.^{1,2} Free testosterone is the form of testosterone that can diffuse into the tissues and act on receptors and is considered the active fraction by many physicians.^{1,3,4} Free testosterone assessment is recommended as a primary or secondary measure of androgen activity in men^{2,5-11} and women.¹²⁻¹⁶ The equilibrium dialysis or ultrafiltration methods are recommended, however they are relatively difficult and expensive to perform. All of these methods are currently available from LabCorp.

The primary screening test for the diagnosis of hypoandrogenism in men is the measurement of total testosterone in serum in a morning sample.^{2,5-11} Low concentrations of testosterone in serum should be confirmed by repeat measurement, preferably using liquid chromatography/mass spectrometry.^{6,7,10} Determination of free testosterone can be of value in men with borderline total testosterone because alterations in SHBG levels can markedly affect the concentration of biologically available free testosterone.^{2,17-20} Factors and conditions that tend to increase SHBG concentrations in men include aging, hyperthyroidism, estrogens, HIV disease, anticonvulsant therapy, and liver disease.² Factors and conditions that tend to decrease SHBG concentrations in men include obesity, diabetes mellitus, hypothyroidism, glucocorticoid therapy, androgenic steroid therapy, nephrotic syndrome and acromegaly.²

The binding affinities between testosterone, SHBG and albumin have been measured experimentally, and it has been shown that if the concentrations of serum total testosterone, SHBG, and albumin are measured, the corresponding concentration of free testosterone can be accurately estimated according to equilibrium kinetics.²¹ The widely accepted method of calculation of serum free testosterone is based upon a 1999 paper by Vermeulen.²¹ The authors derived the equation based upon the law of mass action. Using the known association constants between testosterone, SHBG, and albumin, they calculated the theoretically predicted serum free testosterone concentrations. This method demonstrated substantial agreement with equilibrium dialysis measured concentrations when testosterone, SHBG and albumin are measured using reliable assays.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

See individual test components.

Specimen Requirements

Specimen

Serum

Volume

1.5 mL

Minimum Volume

1 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube or gel-barrier tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.²²

Testosterone, Free, Calc		
Age	Male (pg/mL)	Female (pg/mL)
0 to 1 y	Not established	Not established
2 to 5 y	0.1–0.5	0.2–0.7
6 to 7 y	0.1–0.9	0.4–7.6
8 y	0.2–0.8	0.4–7.6
9 y	0.2–1.4	0.4–7.6
10 y	0.3–3.6	0.4–7.6
11 y	0.3–7.9	0.4–7.6
12 y	0.6–32.0	0.4–7.6
13 y	1.4–98.1	1.6–15.4
14 y	8.5–105.6	1.6–15.4
15 y	14.0–128.1	1.6–15.4
16 y	30.5–147.6	1.6–15.4
17 y	38.7–157.6	1.6–15.4
18 to 30 y	47.7–173.9	1.1–12.9
31 to 40 y	42.3–190.0	0.7–7.9
41 to 50 y	30.3–183.2	0.5–6.0

Testosterone, Free, Calc		
Age	Male (pg/mL)	Female (pg/mL)
51 to 60 y	35.8–168.2	0.4–6.7
61 to 70 y	34.7–150.3	0.4–7.0
71 to 80 y	31.7–120.8	0.3–5.7
81 to 100 y	20.7–97.4	0.2–5.6

Storage Instructions

Room temperature

Causes for Rejection

See individual test components.

Footnotes

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